MINI SOC LAB

Complete Setup Guide

ELK Stack | *Snort IDS* | *TheHive* | *MITRE ATT&CK*

Project 1 — Mini SOC Logical Design

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1. Lab Overview & Architecture

This guide documents the complete setup of a Mini Security Operations Center (SOC) lab built on VMware. The lab demonstrates a 5-layer security monitoring architecture using open-source tools.

1.1 SOC Architecture

Layer	Components & Purpose
Layer 1 — Log Sources	Windows 10, Ubuntu, Snort IDS, Web Server (Flask)
Layer 2 — Collection	ELK Stack (Elasticsearch, Logstash, Kibana), Winlogbeat, Filebeat
Layer 3 — SIEM Core	Kibana Dashboards, Detection Rules, MITRE ATT&CK Mapping
Layer 4 — IR	TheHive Case Management, IOC Analysis, Triage
Layer 5 — Reporting	KPI Dashboards, Incident Reports, Root Cause Analysis

1.2 Network Topology

Machine	IP Address / Details
Ubuntu ELK Server	192.168.1.127 (ens33), Tailscale: 100.89.152.37
Windows 10 Workstation	192.168.1.129 (GROUP2-Ali-Mohamed)
OPNsense Firewall	LAN: 192.168.79.1 (VMnet1)
Snort IDS	Running on Ubuntu — monitoring ens33
Flask Web Server	Running on Ubuntu — port 5000

2. Environment & Prerequisites

2.1 Software Requirements

Software	Version / Notes
Ubuntu Server	24.04 LTS (VMware VM)
Docker	Latest stable
Docker Compose	v2+
Node.js	For website (optional)
Python 3	For Flask website
Winlogbeat	9.1.4 (Windows)
Filebeat	9.1.4 (Ubuntu)
ELK Stack	9.3.0 (via docker-elk)
TheHive	4.1.24 (Docker)

2.2 VMware Network Setup

- Network Adapter 1 (ens33): NAT or Bridged → 192.168.1.x
- Network Adapter 2 (VMnet1): Host-only → 192.168.79.x (for OPNsense)
- Both adapters type: E1000

2.3 Credentials Reference

Service	Credentials
Kibana / Elasticsearch	elastic / depi123
TheHive Admin	admin@thehive.local / secret
TheHive Analyst	analyst@soc.lab / analyst123
OPNsense	root / opnsense
Winlogbeat	No auth — connects to port 5044

3. Layer 1 — Log Sources

This layer defines all the machines and services that generate security-relevant logs.

Log Source	Status / Notes
Windows 10 (Security Events)	✅ Active — via Winlogbeat
Ubuntu Syslog & Auth	✅ Active — via Filebeat
Snort IDS Alerts	✅ Active — via Filebeat
Flask Web Server	✅ Active — via Filebeat
OPNsense Firewall Syslog	⏸ Pending — requires LAN connectivity

3.1 Key Windows Event IDs

Event ID	Description
4624	Successful logon
4625	Failed logon (Brute Force indicator)
4720	New user account created
4672	Special privileges assigned
4689	Process termination
4732	User added to group

4. Layer 2 — Log Collection (ELK Stack)

4.1 ELK Stack Installation

The ELK Stack is deployed using the deviantony/docker-elk project on Docker.

Step 1 — Clone the repository:

```
git clone https://github.com/deviantony/docker-elk.git
cd docker-elk
```

Step 2 — Set passwords in .env file:

```
ELASTIC_PASSWORD=depi123
LOGSTASH_INTERNAL_PASSWORD=depi123
KIBANA_SYSTEM_PASSWORD=depi123
```

Step 3 — Run setup and start:

```
docker compose up setup
docker compose up -d
```

Step 4 — Generate Kibana encryption keys:

```
docker compose exec kibana bin/kibana-encryption-keys generate
```

Add the generated keys to kibana/config/kibana.yml:

```
xpack.encryptedSavedObjects.encryptionKey: <generated>
xpack.reporting.encryptionKey: <generated>
xpack.security.encryptionKey: <generated>
```

> Access Kibana at: http://192.168.1.127:5601 — Login: elastic / depi123

4.2 Fix logstash_internal Permissions

Run these commands to grant Logstash write permissions:

```
# Grant role permissions
curl -u elastic:depi123 -X PUT
"http://localhost:9200/_security/role/logstash_writer" \
  -H "Content-Type: application/json" \
  -d
'{"cluster":["manage_index_templates","monitor"],"indices":[{"names":["*"],"pri
vileges":["write","create","create_index","manage","auto_configure"]}]}'

# Assign role to user
curl -u elastic:depi123 -X PUT
"http://localhost:9200/_security/user/logstash_internal" \
  -H "Content-Type: application/json" \
```

```
  -d '{"password":"depi123","roles":["logstash_writer"],"full_name":"Logstash
Internal"}'

# Restart Logstash
cd ~/docker-elk && docker compose restart logstash
```

4.3 Logstash Pipeline Configuration

File: ~/docker-elk/logstash/pipeline/logstash.conf

```
input {
  beats { port => 5044 }
  udp { port => 5140; type => "pfsense" }
}
filter {
  if [@metadata][beat] == "winlogbeat" or [agent][type] == "winlogbeat" {
    mutate { add_field => { "source" => "windows" } }
  }
  else if [fields][source] == "snort" {
    mutate { add_field => { "source" => "snort" } }
  }
  else if [type] == "pfsense" {
    mutate { add_field => { "source" => "pfsense" } }
  }
  else if [fields][source] == "linux" {
    mutate { add_field => { "source" => "linux" } }
  }
}
output {
  if [source] == "windows" {
    elasticsearch { hosts=>["elasticsearch:9200"] index=>"windows-logs-
%{+YYYY.MM.dd}"
      user=>"logstash_internal" password=>"${LOGSTASH_INTERNAL_PASSWORD}" }
  }
  else if [source] == "snort" {
    elasticsearch { hosts=>["elasticsearch:9200"] index=>"snort-logs-
%{+YYYY.MM.dd}"
      user=>"logstash_internal" password=>"${LOGSTASH_INTERNAL_PASSWORD}" }
  }
  else {
    elasticsearch { hosts=>["elasticsearch:9200"] index=>"generic-logs-
%{+YYYY.MM.dd}"
      user=>"logstash_internal" password=>"${LOGSTASH_INTERNAL_PASSWORD}" }
  }
}
```

4.4 Winlogbeat Setup (Windows)

Download Winlogbeat 9.1.4 from elastic.co and install on Windows 10.

Configuration file (winlogbeat.yml):

```
winlogbeat.event_logs:
  - name: Security
  - name: System
  - name: Application
output.logstash:
  hosts: ["192.168.1.127:5044"]
setup.kibana:
  host: "192.168.1.127:5601"
```

Install and start the service (PowerShell as Administrator):

```
PowerShell -ExecutionPolicy Bypass -File .\install-service-winlogbeat.ps1
Start-Service winlogbeat
```

Verify logs flowing to ELK:

```
curl -u elastic:depi123 "http://localhost:9200/_cat/indices?v" | grep windows
```

4.5 Snort IDS Setup

Snort is installed on the Ubuntu ELK server to monitor network traffic on ens33.

Installation:

```
sudo apt install snort -y
# During install: interface = ens33, network = 192.168.1.0/24
```

Add a custom ICMP detection rule:

```
sudo nano /etc/snort/rules/local.rules

# Add this line:
alert icmp any any -> any any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)
sudo service snort restart
sudo service snort status
```

4.6 Filebeat Setup (Ubuntu)

Filebeat ships Snort alerts and system logs to Logstash.

Install Filebeat:

```
wget https://artifacts.elastic.co/downloads/beats/filebeat/filebeat-9.1.4-
amd64.deb
sudo dpkg -i filebeat-9.1.4-amd64.deb
```

Configuration (/etc/filebeat/filebeat.yml):

```
filebeat.inputs:
- type: filestream
  enabled: true
  id: snort-alerts
  paths:
  - /var/log/snort/snort.alert.fast
  fields:
    source: snort
  fields_under_root: true
- type: filestream
  enabled: true
  id: linux-logs
  paths:
  - /var/log/syslog
  - /var/log/auth.log
  fields:
    source: linux
  fields_under_root: true
- type: filestream
  enabled: true
  id: webapp-logs
  paths:
  - /var/log/webapp/access.log
  fields:
    source: webapp
  fields_under_root: true

output.logstash:
  hosts: ["192.168.1.127:5044"]
sudo systemctl enable filebeat
sudo systemctl start filebeat
```

5. Layer 3 — SIEM Core (Kibana)

5.1 Create Kibana Data Views

Go to: Stack Management → Kibana → Data Views → Create data view

Name	Index Pattern
Windows Logs	windows-logs-*
Snort Logs	snort-logs-*
Linux Logs	linux-logs-*
Generic Logs	generic-logs-*

5.2 Windows Security Dashboard

Navigate to: Dashboards → Create dashboard → Create visualization

Visualization	Type / Field
Windows Events by ID	Bar chart — event.code.keyword
Windows Events Timeline	Area chart — @timestamp
Events by Host	Bar chart — host.name.keyword
Failed Login Attempts	Bar chart — filter: event.code:4625
Audit Success vs Failure	Pie chart — winlog.keywords.keyword

5.3 Detection Rules

Navigate to: Security → Rules → Detection rules (SIEM) → Create new rule

Rule Name	KQL Query / Details
Brute Force Login Attempt	event.code: "4625" | Threshold: 5 in 60s | Severity: High
New User Account Created	event.code: "4720" | Severity: Critical
Special Privileges Assigned	event.code: "4672" | Severity: High
ICMP Flood Detected	message: "ICMP Ping Detected" | Index: snort-logs-* | Medium

5.4 MITRE ATT&CK Mapping

Edit each rule → About tab → Advanced settings → MITRE ATT&CK threats

Detection Rule	MITRE Technique
Brute Force Login	T1110 — Brute Force (Credential Access)
New User Created	T1136 — Create Account (Persistence)
Special Privileges	T1087 — Account Discovery (Discovery)
ICMP Flood	T1046 — Network Service Discovery (Discovery)

6. Layer 4 — Incident Response (TheHive)

6.1 TheHive Installation

Create the docker-compose.yml file:

```
mkdir ~/thehive && cd ~/thehive
nano docker-compose.yml
```

docker-compose.yml content:

```
services:
  elasticsearch:
    image: elasticsearch:7.17.9
    container_name: thehive-es
    restart: unless-stopped
    environment:
      - discovery.type=single-node
      - xpack.security.enabled=false
      - ES_JAVA_OPTS=-Xms512m -Xmx512m
    volumes:
      - thehive-es-data:/usr/share/elasticsearch/data
  thehive:
    image: thehiveproject/thehive4:latest
    container_name: thehive
    restart: unless-stopped
    depends_on:
      - elasticsearch
    ports:
      - "9000:9000"
    environment:
      - ES_HOSTNAME=elasticsearch
      - SECRET=mySecretKey1234
volumes:
  thehive-es-data:
```

Start TheHive:

```
docker compose up -d
docker compose ps
```

> Access TheHive at: http://192.168.1.127:9000 — Login: admin@thehive.local / secret

6.2 Organisation & User Setup

Step 1 — Login as admin and create SOC organisation:

- Go to: Admin menu → Organisations → + New Organisation
- Name: SOC-Lab | Description: SOC Lab Organisation

Step 2 — Create SOC Analyst user:

- Click SOC-Lab → Users tab → + Create new user
- Login: analyst@soc.lab | Name: SOC Analyst | Profile: analyst
- Set password: analyst123

6.3 Creating a Case (Incident Response Workflow)

Login as analyst@soc.lab and create a case:

Field	Example Value
Title	Brute Force Attack Detected
Severity	High
TLP	Amber
Tags	brute-force, windows, T1110
Description	Multiple failed login attempts from Windows host

Add tasks to the case:

- Task 1: Investigate source IP
- Task 2: Check Windows Event Logs in Kibana
- Task 3: Block attacker IP (Containment)
- Task 4: Document findings and close case

7. Layer 5 — Reporting & KPIs

7.1 KPI Metrics Dashboard in Kibana

Create a new dashboard named 'SOC KPI Dashboard' with the following visualizations:

KPI Metric	How to Create
Total Windows Events	Metric viz — windows-logs-* — Count
Failed Login Attempts	Metric viz — filter: event.code:4625
Snort Alert Count	Metric viz — snort-logs-* — Count
Alerts by Severity	Pie chart — .alerts-security* — severity
Events Timeline	Area chart — @timestamp over time
Top Event Types	Horizontal bar — event.action.keyword

7.2 SOC Kibana Space

Create a dedicated Kibana Space for the SOC:

- Go to: Stack Management → Kibana → Spaces → Create a space
- Name: SOC Operations | URL: soc-operations
- Enable: Discover, Dashboard, Security, Alerts, Canvas
- Move all SOC dashboards into this space

7.3 Key KPI Definitions

KPI	Definition
MTTD	Mean Time to Detect — time from attack start to first alert
MTTR	Mean Time to Respond — time from alert to containment
False Positive Rate	% of alerts that are not real threats
Alert Volume	Total number of alerts per time period
Case Resolution Rate	% of TheHive cases resolved vs open

8. Flask Web Server (E-commerce App)

An intentionally vulnerable Flask e-commerce application is deployed as an additional log source.

8.1 Deploy the Web Server

```
cd ~/Desktop/website-main
pip3 install flask --break-system-packages

# Edit app.py to bind to all interfaces and enable logging
# Change last line to:
# app.run(host='0.0.0.0', port=5000, debug=True)

# Add logging after app = Flask(__name__):
# import logging
# logging.basicConfig(
#     filename='/var/log/webapp/access.log',
#     level=logging.INFO,
#     format='%(asctime)s %(levelname)s %(message)s'
# )
```

```
sudo mkdir -p /var/log/webapp
sudo chmod 777 /var/log/webapp
python3 app.py
```

> Access the web app at: http://192.168.1.127:5000

8.2 Website Features

- User registration and login
- Product catalog with filtering
- Shopping cart and checkout
- Admin panel (product management, user management, orders)
- SQLite database (store.db)

9. Troubleshooting Reference

9.1 ELK Stack Issues

Issue	Solution
Kibana not accessible	docker compose restart kibana — wait 2 minutes
logstash_internal 403 error	Re-run the role/user curl commands in section 4.2
No logs in Elasticsearch	Check: docker compose logs logstash --tail=20
Yellow index status	curl -u elastic:depi123 -X PUT "http://localhost:9200/_settings" -H "Content-Type: application/json" -d '{"index":{"number_of_replicas":0}}'

9.2 Filebeat Issues

Issue	Solution
Filebeat fails to start	Check: sudo filebeat -e -c /etc/filebeat/filebeat.yml
'log' input deprecated	Change type from 'log' to 'filestream' and add 'id' field
No snort logs flowing	Verify: ls -la /var/log/snort/ — check snort.alert.fast exists
YAML parse error	Re-edit with nano — ensure correct indentation

9.3 TheHive Issues

Issue	Solution
TheHive keeps restarting	Check: docker compose logs thehive --tail=20
ES connection refused	Ensure thehive-es container is healthy first
Cannot login	Default: admin@thehive.local / secret
Unknown parameter error	Check docker-compose.yml command syntax

9.4 OPNsense Issues

Issue	Solution
Only 1 interface detected	Add second VMware network adapter (E1000 type)
LAN IP not set	Console option 2 — set LAN IP to 192.168.79.1/24
Cannot access web UI	Ensure your browser machine is on VMnet1
WAN no carrier	Change adapter type to Bridged in VMware settings

10. Services & Ports Reference

Service	Port / URL
Kibana	http://192.168.1.127:5601
Elasticsearch	http://192.168.1.127:9200
Logstash Beats Input	192.168.1.127:5044 (TCP)
Logstash Syslog Input	192.168.1.127:5140 (UDP)
TheHive	http://192.168.1.127:9000
Flask Web App	http://192.168.1.127:5000
OPNsense Web UI	http://192.168.79.1
Logstash API	192.168.1.127:9600

10.1 UFW Firewall Rules

```
sudo ufw allow 5044/tcp   # Logstash Beats
sudo ufw allow 5140/udp   # Logstash Syslog
sudo ufw allow 5601/tcp   # Kibana
sudo ufw allow 9200/tcp   # Elasticsearch
sudo ufw allow 9000/tcp   # TheHive
sudo ufw allow 5000/tcp   # Flask Web App
```

10.2 Docker Services Status Check

```
# Check ELK stack
cd ~/docker-elk && docker compose ps

# Check TheHive
cd ~/thehive && docker compose ps

# Check all indices
curl -u elastic:depi123 "http://localhost:9200/_cat/indices?v"

# Check Snort
sudo service snort status

# Check Filebeat
sudo systemctl status filebeat
```